5.1 nth Roots and Rational Exponents

Name _____ Period ____ Date _____

Algebra II 5.1 (30-minute plan) · Student sheet

I can…

- find nth roots of numbers.
- evaluate expressions with rational exponents.
- solve equations using nth roots.

Vocabulary · fill in the missing word

nth root of a — a number b with b^n = _____

Index — _____

Rational exponent — $a^{\frac{1}{n}}$ = _____ and $a^{\frac{m}{n}} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$

Part 1 · Notes and examples (fill in each note, then work the example)

Real nth roots: If n is **odd**, a has exactly _____ real nth root.

Real nth roots: If n is **even** and a > 0, a has _____ real nth roots. If n is even and a < 0, a has _____ real nth roots.

Example 1

Find the indicated real nth root(s) of a.

a. n = 3, a = −125 b. n = 4, a = 16

Rational exponents: $a^{\frac{1}{n}}$ = _____ $a^{\frac{m}{n}}$ = _____ = _____

Negative rational exponents: $a^{-\frac{m}{n}}$ = _____

Example 2

Evaluate each expression.

a. $16^{\frac{1}{4}}$ b. $(-27)^{\frac{1}{3}}$ c. $9^{\frac{3}{2}}$ d. $32^{-\frac{3}{5}}$

Radical form: In $a^{\frac{m}{n}}$, the denominator n is the _____ and the numerator m is the _____.

Example 3

a. Write $7^{\left(\frac{3}{4}\right)}$ in radical form. b. Write $\sqrt[5]{x^2}$ with a rational exponent.

Part 2 · Partner practice

Partner practice

Evaluate $27^{\left(\frac{2}{3}\right)}$ and $64^{\left(-\frac{1}{2}\right)}$.